

Experiment 4 Report: Final Difficulty Tournament

1. Objective

Experiment 4 evaluates the final Easy / Medium / Hard difficulty settings for the Gomoku AI. Unlike Experiment 3, which screened several candidate configurations, Experiment 4 focuses on the final difficulty ladder.

The current folder contains two result sets:

Result folder	Difficulty setup	Purpose
result(A1_A3_B2)	Easy = A1, Medium = A3, Hard = B2	Original final setup from the earlier candidate selection.
result(A1_B2_B3)	Easy = A1, Medium = B2, Hard = B3	Updated setup after the Eval B rerun showed B2/B3 as stronger practical candidates.

The goal is to check whether the difficulty ladder behaves as expected:

- Hard should be strongest overall.
- Medium should be clearly stronger than Easy.
- Easy should remain fast and weaker.
- Runtime should remain acceptable for each difficulty level.

2. AI Configurations

Setup 1: A1 / A3 / B2

Difficulty	Profile	Search	Evaluation	Depth	Candidate radius
Easy	A1	Alpha-Beta	Eval A	1	2
Medium	A3	Alpha-Beta + ordering	Eval A	3	2
Hard	B2	Alpha-Beta + ordering	Eval B	2	3

Setup 2: A1 / B2 / B3

Difficulty	Profile	Search	Evaluation	Depth	Candidate radius
Easy	A1	Alpha-Beta	Eval A	1	2
Medium	B2	Alpha-Beta + ordering	Eval B	2	3
Hard	B3	Alpha-Beta + ordering	Eval B	3	2

The second setup is motivated by the updated Experiment 3 results, where B3 had the highest score rate and B2 had a strong strength-runtime trade-off.

3. Tournament Protocol

Both result sets use:

- 3 difficulty AIs;
- 3 pairings: Easy vs Medium, Easy vs Hard, Medium vs Hard;
- fixed opening positions;
- max moves: 50;
- draw if max moves is reached.

The difference is the number of games:

Result folder	Games per pair	Total games
result(A1_A3_B2)	200	600
result(A1_B2_B3)	50	150

The tournament uses fixed openings rather than a single empty-board start. This reduces overfitting to one deterministic game path and gives each pair a broader set of starting positions.

4. Overall Results

Result folder	Games	Black wins	White wins	Draws	Max-move draws	Avg moves
result(A1_A3_B2)	600	320	270	10	10	17.57
result(A1_B2_B3)	150	95	52	3	3	15.26

Both tournaments have a low draw count. Black has an advantage in both result sets, especially in `result(A1_B2_B3)` , but white still wins enough games that the results are not only first-player outcomes.

5. Setup 1 Results: A1 / A3 / B2

Ranking

AI	Profile	Evaluation	Depth	Games	Wins	Losses	Draws	Win rate	Score rate	Avg time / move ms	Avg nodes / move	Black win rate	White win rate
Hard	B2	Eval B	2	400	280	110	10	0.7	0.7125	3175.6271	431.85	0.75	0.65
Medium	A3	Eval A	3	400	250	140	10	0.625	0.6375	6228.0128	5356.2	0.65	0.6
Easy	A1	Eval A	1	400	60	340	0	0.15	0.15	17.6245	76.49	0.2	0.1

Pairwise Results

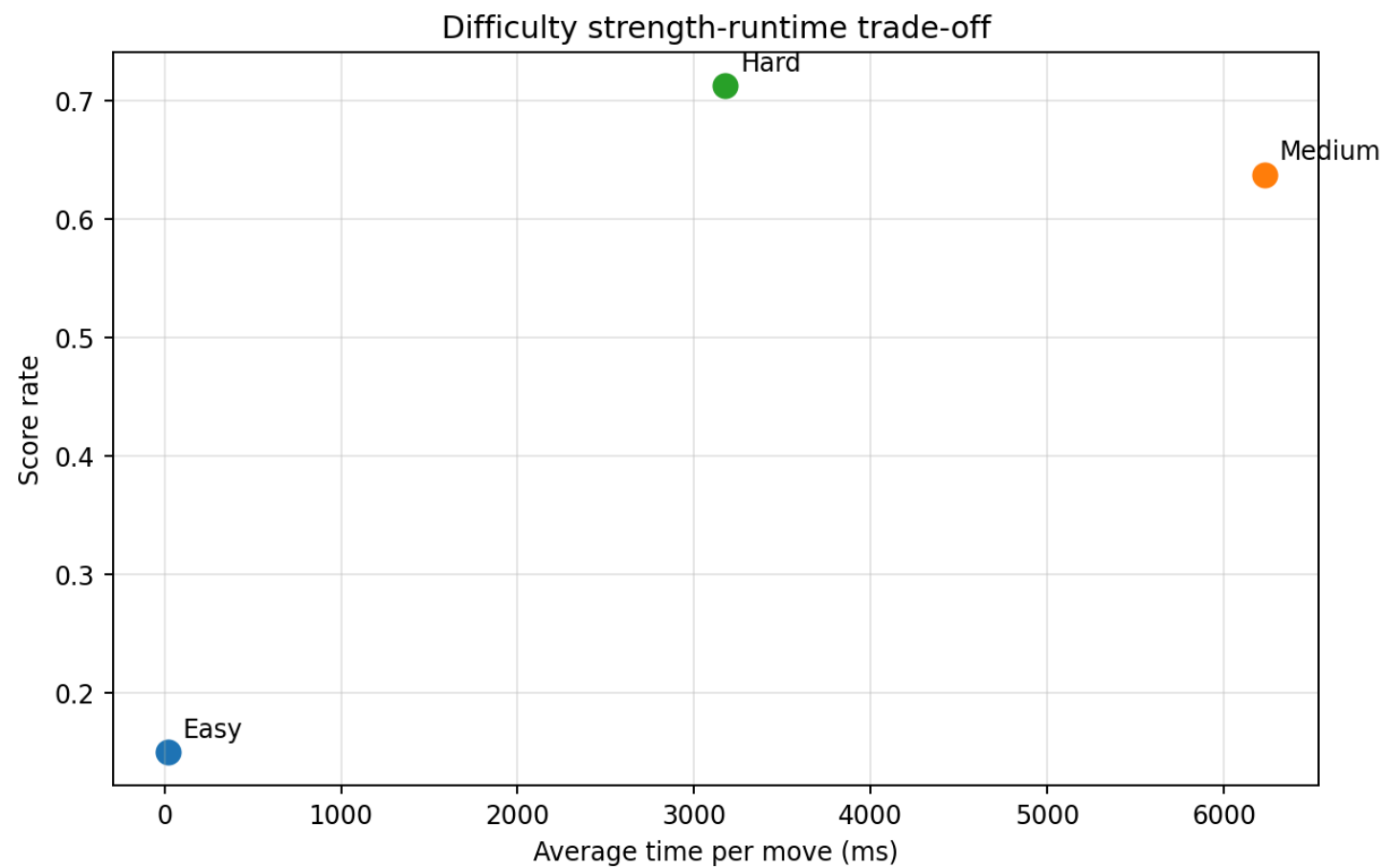
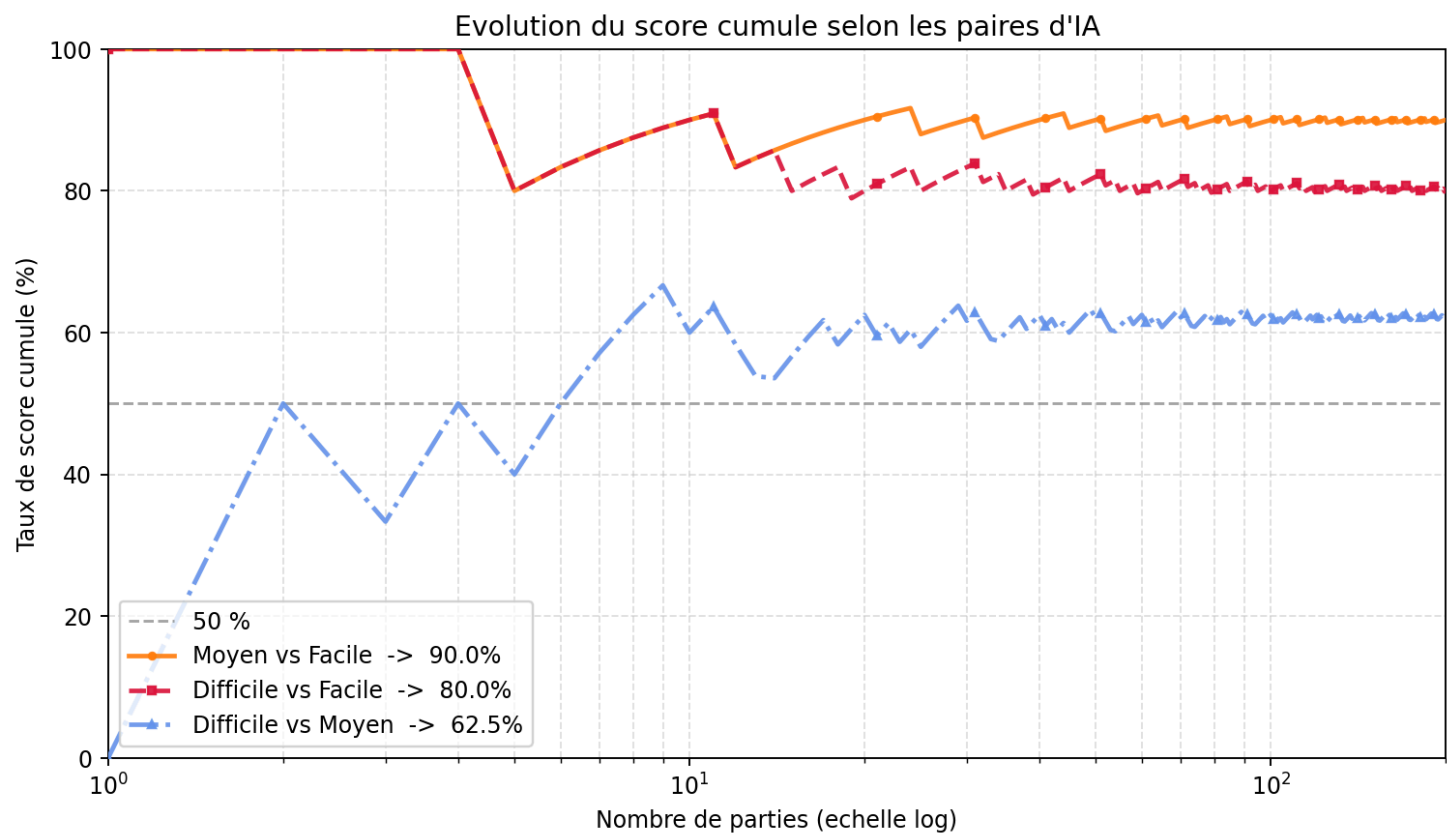
Pair	Games	AI 1 wins	AI 2 wins	Draws	AI 1 score rate	AI 2 score rate	Interpretation
Easy vs Hard	200	40	160	0	0.2	0.8	Hard clearly beats Easy.
Easy vs Medium	200	20	180	0	0.1	0.9	Medium clearly beats Easy.
Medium vs Hard	200	70	120	10	0.375	0.625	Hard beats Medium, but the gap is moderate.

This setup validates the ordering:

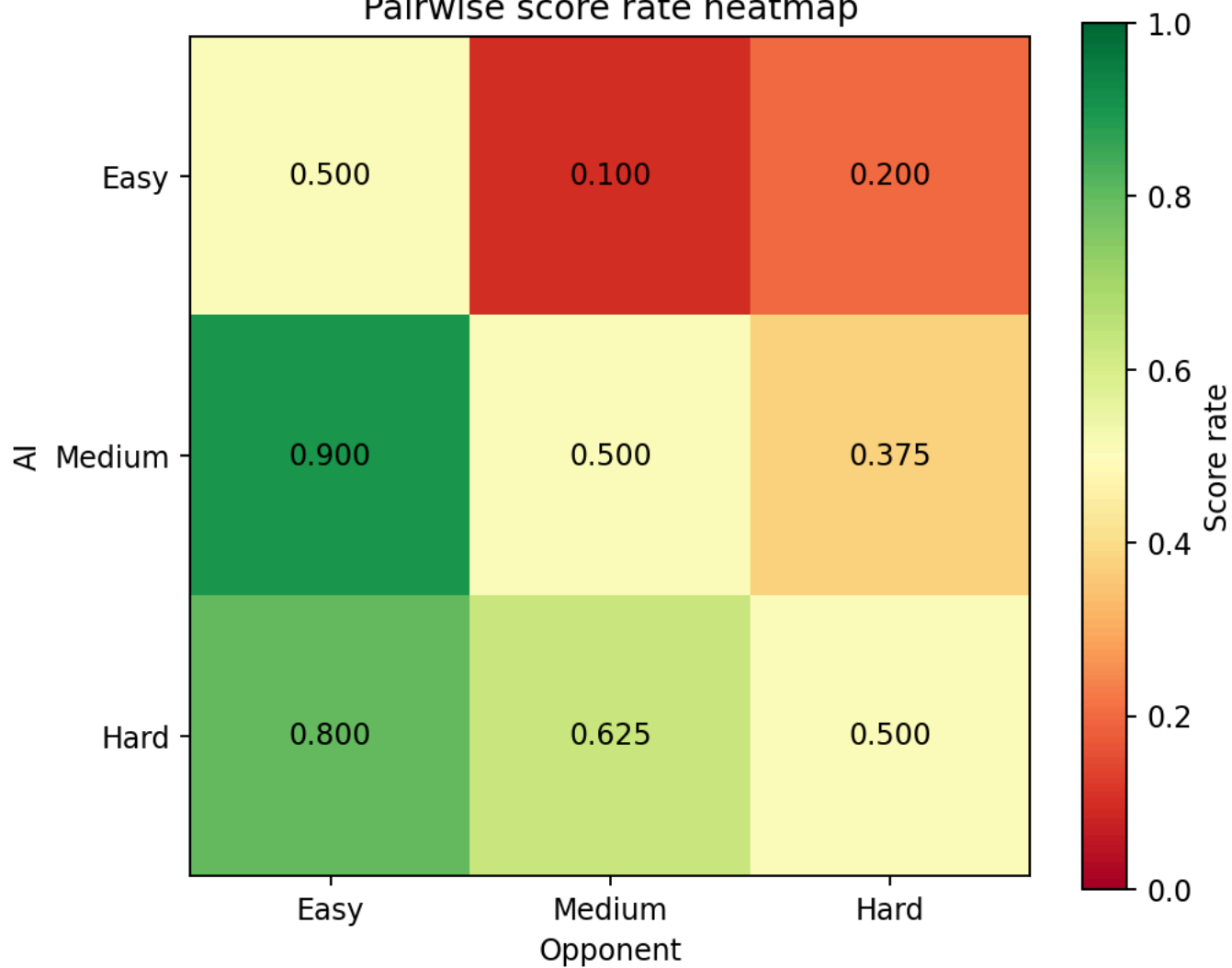
Hard > Medium > Easy

However, Medium is slower than Hard. Medium uses A3, which searches deeper with Eval A, while Hard uses B2, a shallower but more effective Eval B configuration.

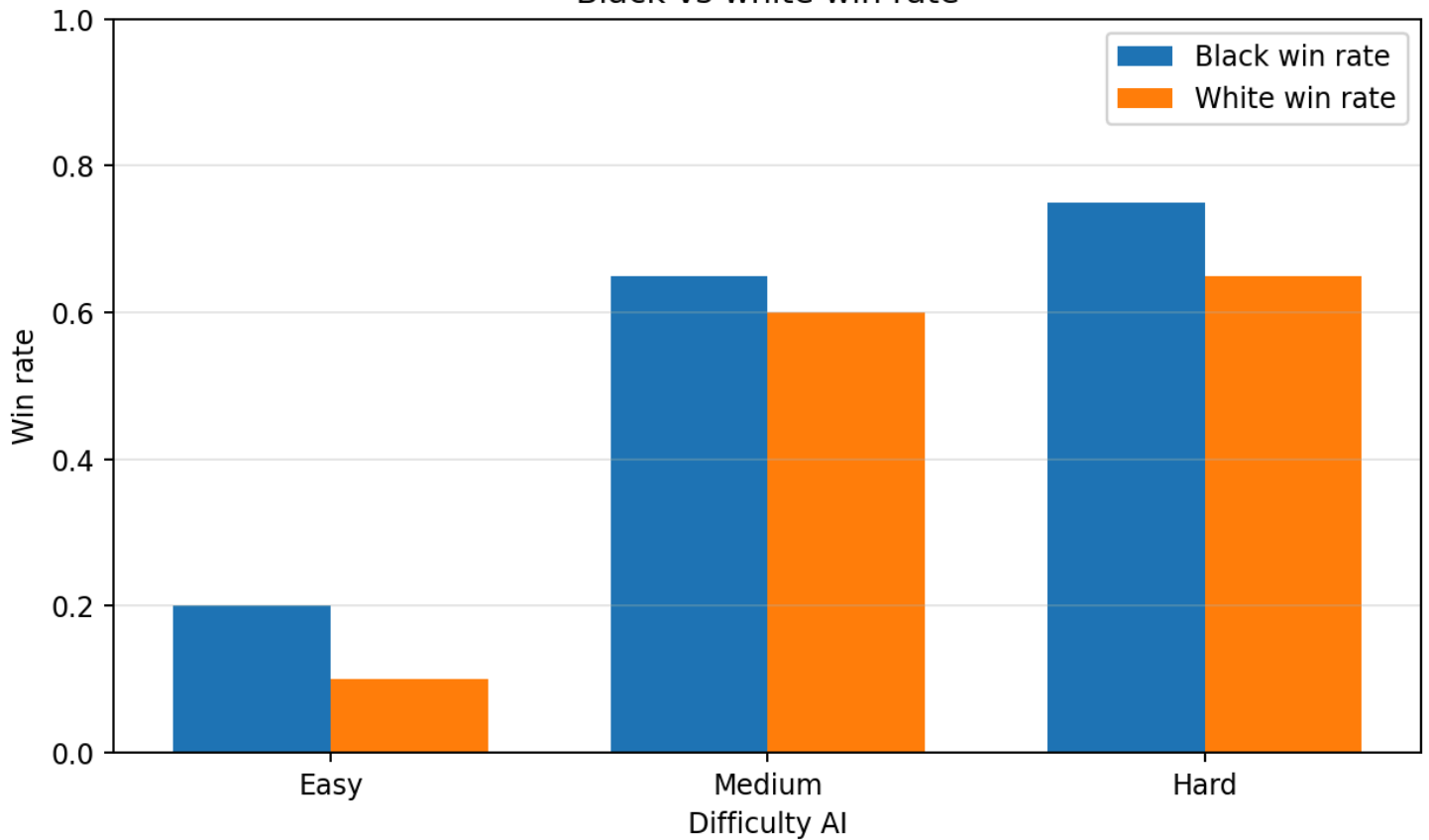
Plots



Pairwise score rate heatmap



Black vs white win rate



6. Setup 2 Results: A1 / B2 / B3

Ranking

AI	Profile	Evaluation	Depth	Games	Wins	Losses	Draws	Win rate	Score rate	Avg time / move ms	Avg nodes / move	Black win rate	White win rate
Hard	B3	Eval B	3	100	65	32	3	0.65	0.665	4843.7375	5126.72	0.8	0.5
Medium	B2	Eval B	2	100	61	36	3	0.61	0.625	3735.4328	427.67	0.76	0.46
Easy	A1	Eval A	1	100	21	79	0	0.21	0.21	21.2439	73.85	0.34	0.08

Pairwise Results

Pair	Games	AI 1 wins	AI 2 wins	Draws	AI 1 score rate	AI 2 score rate	Interpretation
Easy vs Hard	50	12	38	0	0.24	0.76	Hard clearly beats Easy.
Easy vs Medium	50	9	41	0	0.18	0.82	Medium clearly beats Easy.
Medium vs Hard	50	20	27	3	0.43	0.57	Hard beats Medium, but the gap is smaller than in Easy matchups.

This setup also validates the ordering:

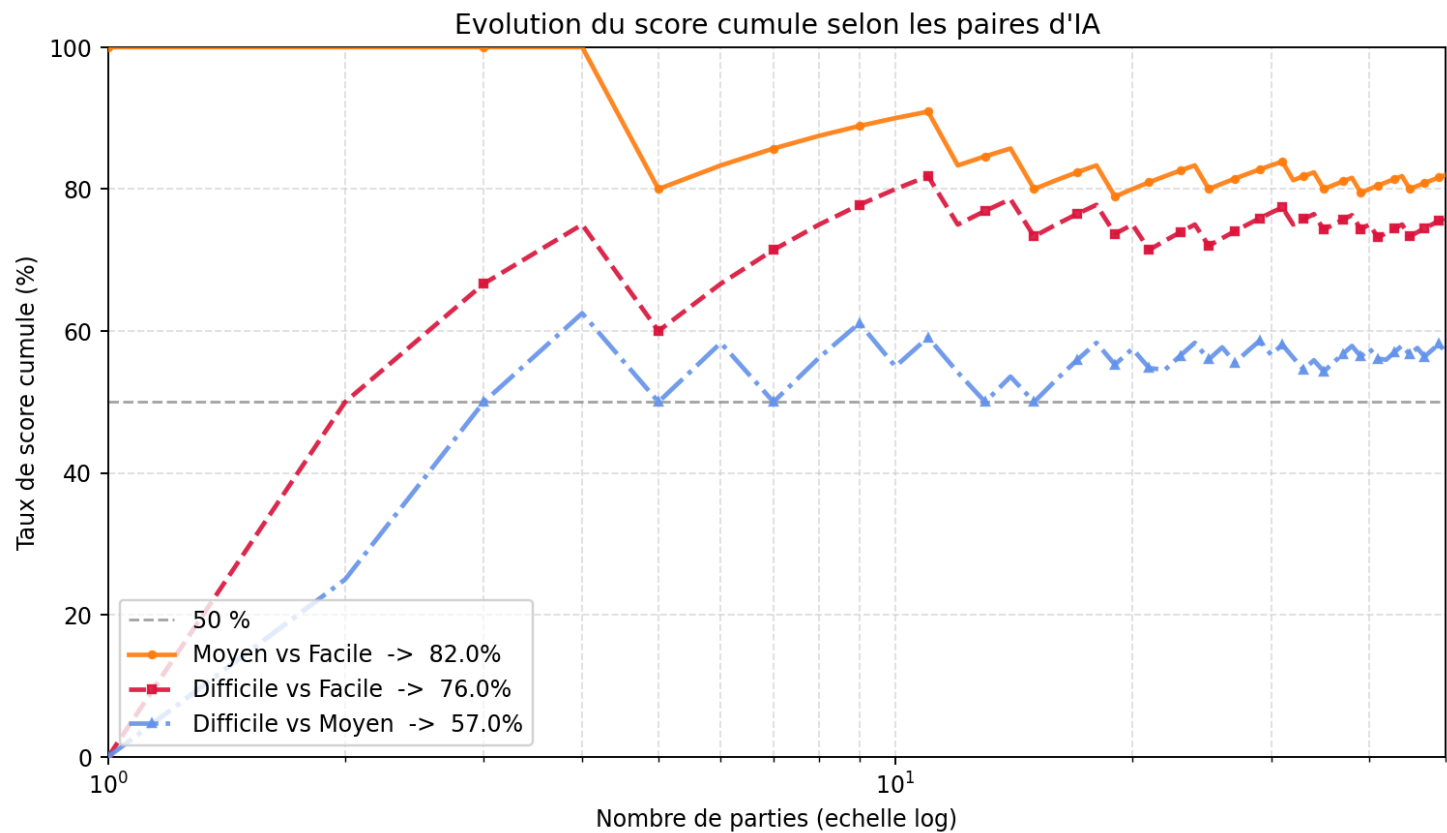
Hard > Medium > Easy

Compared with Setup 1, Setup 2 has a more natural difficulty progression:

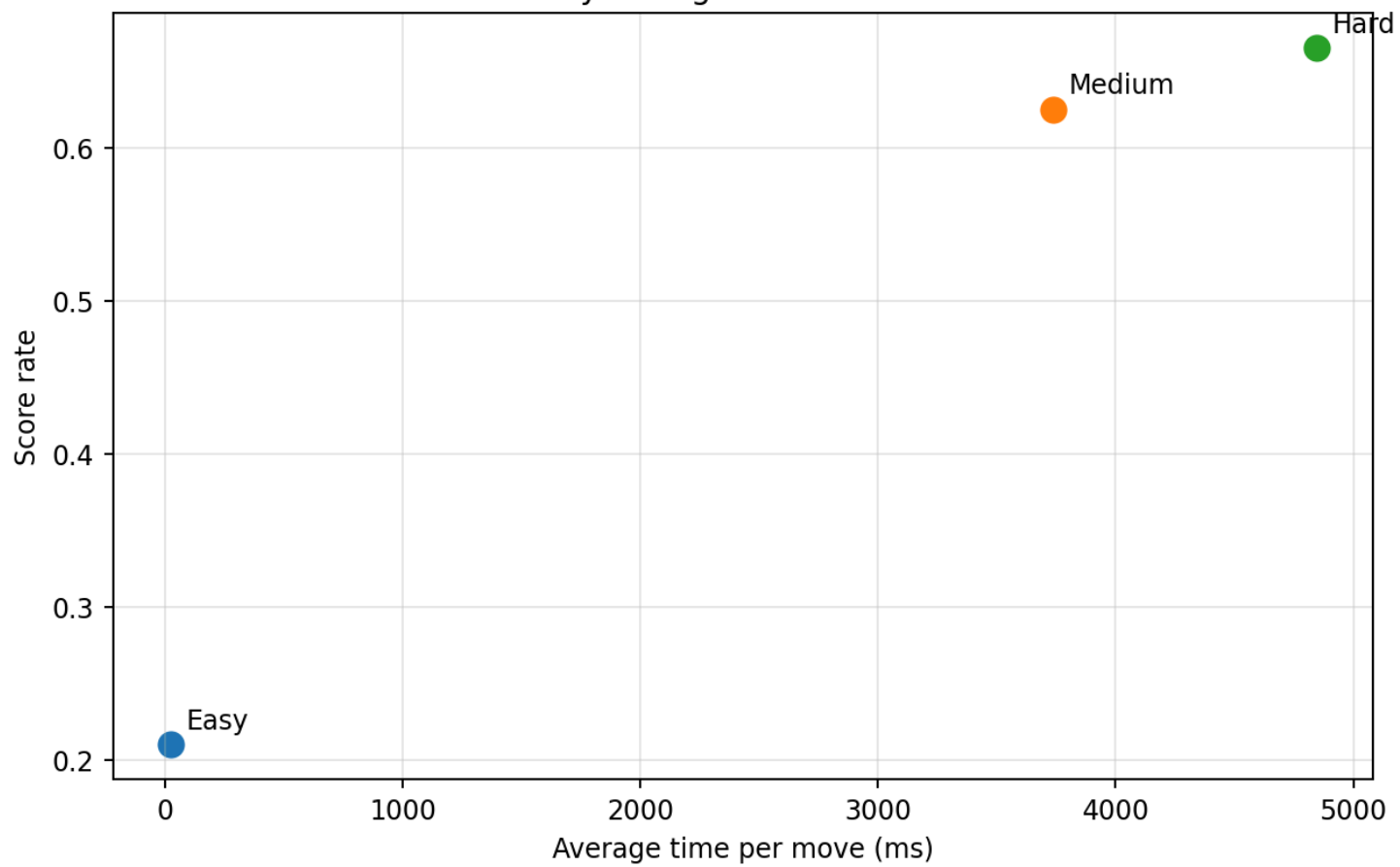
- Easy is still weak and fast.
- Medium is stronger than Easy and uses the efficient B2 configuration.
- Hard is the strongest and uses deeper Eval B search.

The Medium-Hard gap is smaller in Setup 2, which is expected because both levels use Eval B. This makes the final ladder smoother, although it also means Medium and Hard are closer in strength.

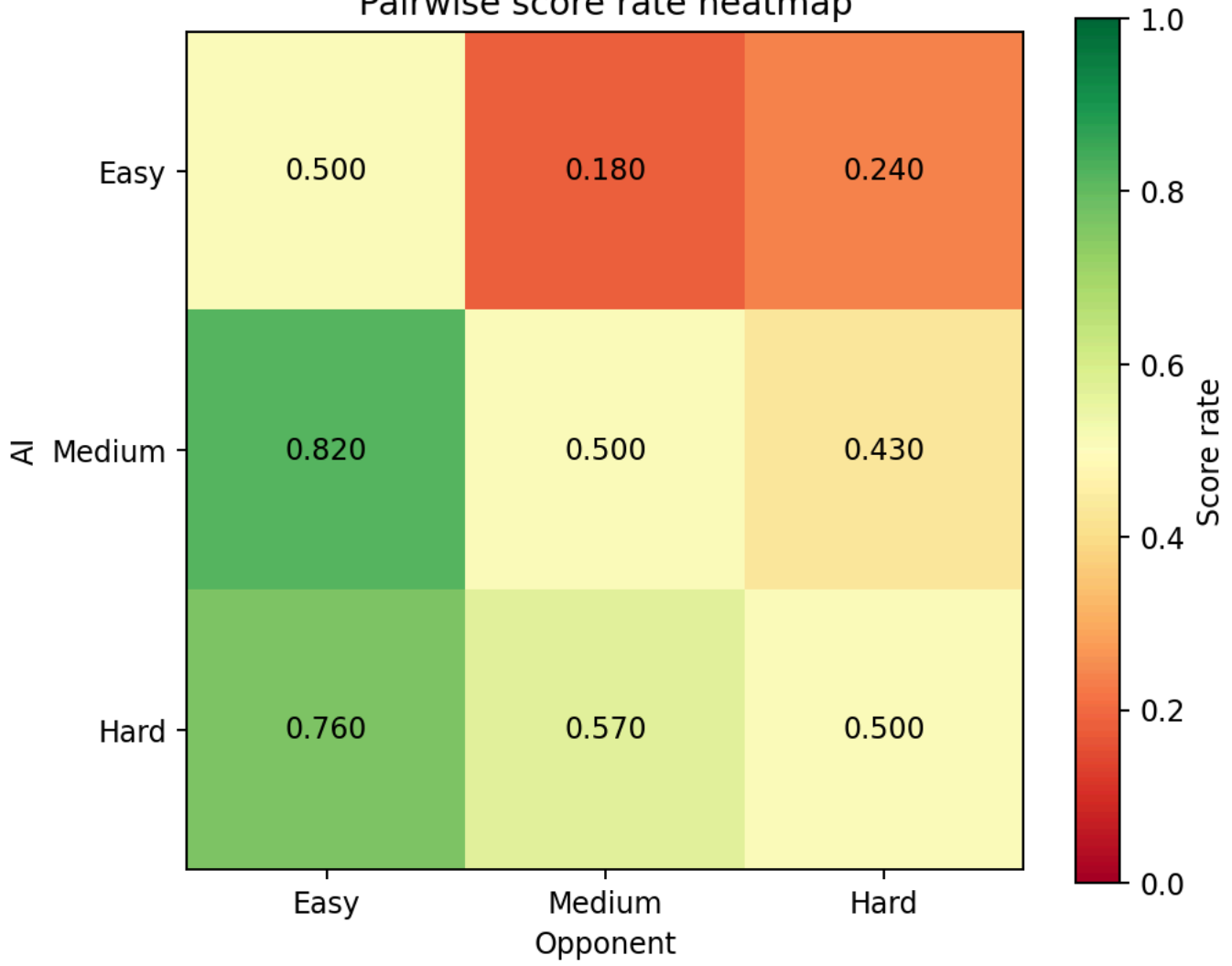
Plots



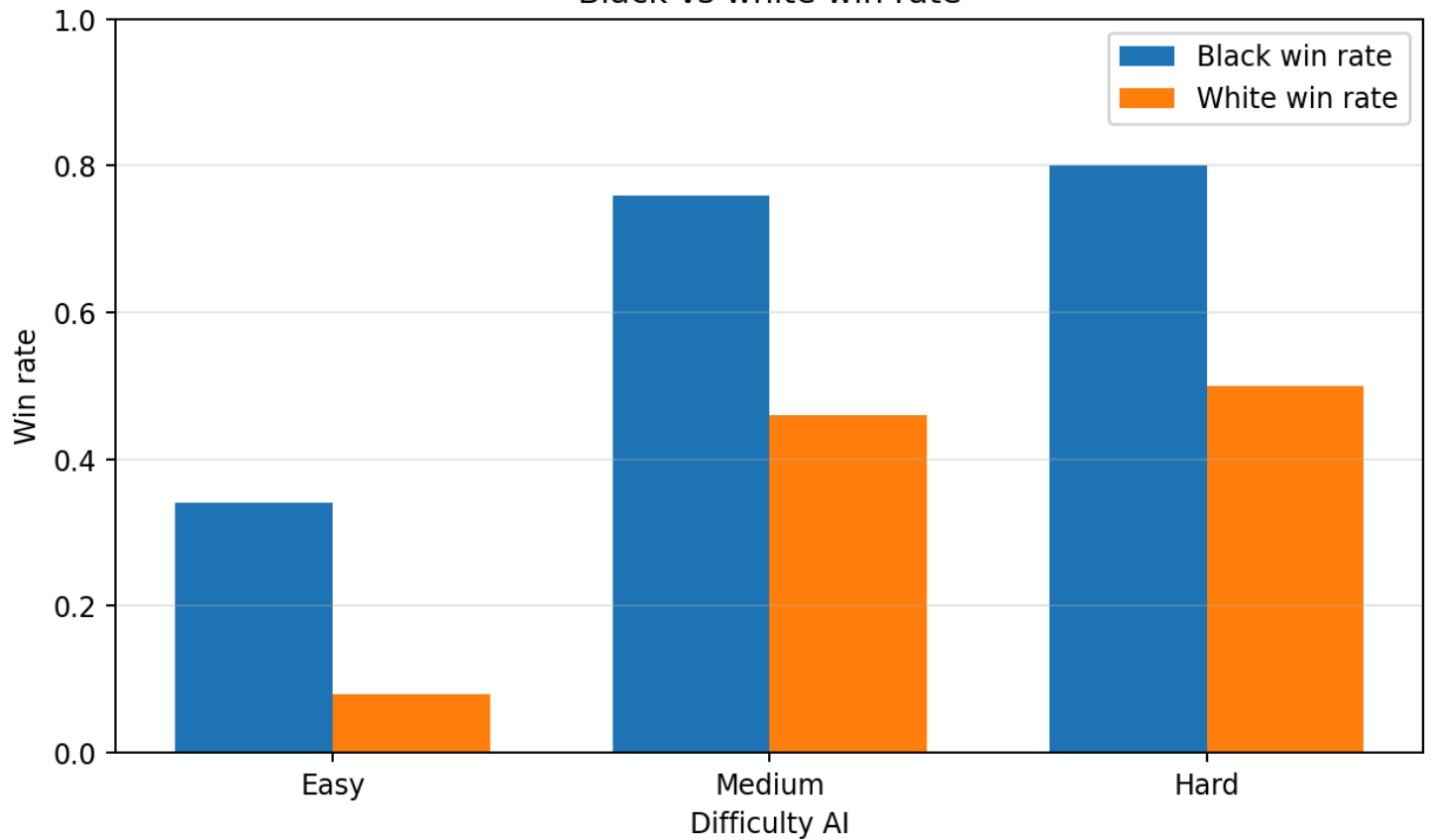
Difficulty strength-runtime trade-off



Pairwise score rate heatmap



Black vs white win rate



7. Comparison Between the Two Setups

Criterion	A1 / A3 / B2	A1 / B2 / B3
Ordering valid?	Yes: Hard > Medium > Easy	Yes: Hard > Medium > Easy
Easy behavior	Weak and fast	Weak and fast
Medium behavior	Strong, but slower than Hard	Strong and more efficient
Hard behavior	Strongest and faster than Medium	Strongest, but slower than Medium
Medium-Hard gap	Hard clearly ahead	Hard ahead, but closer
Runtime logic	Less intuitive because Medium is slowest	More intuitive because deeper B3 is Hard

Setup 1 is statistically stronger because it uses 600 games, while Setup 2 uses 150 games. However, Setup 2 matches the updated Experiment 3 conclusions better: B2 is a strong medium-level candidate and B3 is the strongest candidate among the practical Eval B configurations.

8. First-Player Analysis

Setup 1 color win rates:

```
Easy black win rate:  0.20
Easy white win rate:  0.10
Medium black win rate: 0.65
Medium white win rate: 0.60
Hard black win rate:  0.75
Hard white win rate:  0.65
```

Setup 2 color win rates:

```
Easy black win rate:  0.34
Easy white win rate:  0.08
Medium black win rate: 0.76
Medium white win rate: 0.46
Hard black win rate:   0.80
Hard white win rate:   0.50
```

Black has an advantage in both setups. The advantage is more visible in Setup 2, partly because it has fewer games. This should be mentioned as a limitation when interpreting the 150-game result set.

9. Interpretation

Both result sets support a valid Easy / Medium / Hard ladder. Easy is clearly weaker than the other two levels, which is desirable for a beginner-level AI.

The main difference is the Medium configuration:

- In Setup 1, Medium = A3. It is stronger than Easy but slower than Hard.
- In Setup 2, Medium = B2. It is stronger than Easy, more efficient than A3, and aligns with the updated Experiment 3 candidate ranking.

The Hard configuration also differs:

- In Setup 1, Hard = B2. It is strongest in that setup and efficient.
- In Setup 2, Hard = B3. It is strongest in the updated setup, but more expensive than B2.

Overall, Setup 2 is more coherent with the updated candidate tournament:

```
Easy   = A1
Medium = B2
Hard   = B3
```

10. Conclusion

The recommended final difficulty setup is:

```
Easy   = A1
Medium = B2
Hard   = B3
```

This recommendation is based on the updated Eval B results and the second Experiment 4 result set. It preserves a clear difficulty order while assigning the strongest practical configuration, B3, to Hard.

Important limitation: `result(A1_B2_B3)` has 150 games, while `result(A1_A3_B2)` has 600 games. If more time is available, the A1/B2/B3 setup should be rerun with 200 games per pair to match the sample size of the earlier setup.

11. How to Reproduce

Run the tournament:

```
python experiments/experiment4_difficulty_tournament/tournament.py
```

Regenerate plots:

```
python experiments/experiment4_difficulty_tournament/plot_difficulty_tournament.py
```

Current result folders:

```
experiments/experiment4_difficulty_tournament/result(A1_A3_B2)  
experiments/experiment4_difficulty_tournament/result(A1_B2_B3)
```